

Gaussian Approximation of Token Frequencies for Hidden Markov Models

Research Notes

March 2026

1 Setup

Consider a Hidden Markov Model generating sequences $\mathbf{x} = (x_1, \dots, x_L)$ of length L over a vocabulary of size d . For each sequence, define the *empirical frequency vector*

$$f_j(\mathbf{x}) = \frac{1}{L} \sum_{t=1}^L \mathbf{1}[x_t = j], \quad j = 1, \dots, d, \quad (1)$$

and the deviation from the stationary mean $\delta \mathbf{f} = \mathbf{f} - \mu$, where $\mu_j = \mathbb{E}[f_j]$ under the stationary distribution.

By the central limit theorem for functionals of HMMs, $\sqrt{L} \delta \mathbf{f}$ converges in distribution to a multivariate Gaussian:

$$\sqrt{L} (\mathbf{f} - \mu) \xrightarrow{d} \mathcal{N}(0, \Sigma_{\text{true}}), \quad (2)$$

where Σ_{true} is the asymptotic covariance matrix (independent of L). In particular, $\text{Var}(\mathbf{f}) \approx \Sigma_{\text{true}}/L$ for large L .

Goal. We verify this CLT numerically for the Mess3 process ($d = 3$ tokens, 3 hidden states, uniform stationary token distribution $\mu = (1/3, 1/3, 1/3)$), and extract the inverse covariance $\Sigma_{\text{true}}^{-1}$ from two independent estimates that should agree as $L \rightarrow \infty$.

2 Methodology

For each sequence length L , we generate N sequences from the Mess3 HMM. For each sequence $\mathbf{x}_i$, we compute:

1. The exact log-likelihood $\log P(\mathbf{x}_i)$ via the forward algorithm.
2. The empirical frequency vector $\mathbf{f}_i$ and its deviation $\delta \mathbf{f}_i = \mathbf{f}_i - \mu$.

2.1 Types and the histogram estimator

The *type* of a sequence is its frequency vector. All sequences sharing the same type form a type class. The probability of a type is

$$P(\text{type } \mathbf{f}) = \sum_{\mathbf{x}: \text{freq}(\mathbf{x})=\mathbf{f}} P(\mathbf{x}). \quad (3)$$

If the CLT (??) holds, the type distribution is approximately Gaussian:

$$\log P(\text{type } \mathbf{f}) \approx \text{const} - \frac{L}{2} \delta \mathbf{f}^T \Sigma_{\text{true}}^{-1} \delta \mathbf{f}. \quad (4)$$

Since $\mathbf{f}$ lies on the $(d-1)$ -simplex, $\delta \mathbf{f}$ has $d-1 = 2$ independent degrees of freedom. We project to 2D by dropping the last component.

Estimate 1: Empirical covariance. From N sampled frequency vectors, we compute the sample covariance $\Sigma_{\text{emp}} = \text{Cov}(\delta \mathbf{f})$ and estimate $\Sigma_{\text{true}} \approx L \cdot \Sigma_{\text{emp}}$, giving $\Sigma_{\text{trueemp}}^{-1} = (L \cdot \Sigma_{\text{emp}})^{-1}$.

Estimate 2: Quadratic fit to type probabilities. We estimate $P(\text{type } \mathbf{f})$ via the histogram: group the N sequences by type and set $\hat{P}(\text{type } \mathbf{f}) = \text{count}_{\mathbf{f}}/N$. We then fit a weighted quadratic regression of $\log(\text{count}_{\mathbf{f}}/N)$ on the CLT-scaled variable $\mathbf{z} = \sqrt{L} \delta \mathbf{f}$:

$$\log \hat{P}(\text{type}) \approx a + \mathbf{b}^T \mathbf{z} + \frac{1}{2} \mathbf{z}^T J_z \mathbf{z}, \quad (5)$$

weighted by the count in each type. The fitted Hessian gives $\Sigma_{\text{truefit}}^{-1} = -J_z$.

If the Gaussian approximation holds, these two estimates should agree: $\Sigma_{\text{truefit}}^{-1} \rightarrow \Sigma_{\text{trueemp}}^{-1}$ as $L \rightarrow \infty$.

2.2 Sequence-level analysis

One might also try to approximate $\log P(\mathbf{x})$ itself (not the type probability) as a quadratic in $\delta \mathbf{f}$. We briefly report this for completeness, but it has two fundamental limitations:

1. **Ordering noise.** For an HMM, $\log P(\mathbf{x})$ depends on the full sequence ordering, not just the frequency vector. This within-type variance is $O(\sqrt{L})$ in standard deviation, while the total variance of $\log P(\mathbf{x})$ grows as $O(L)$. The fraction explainable by a quadratic in $\delta \mathbf{f}$ therefore shrinks: $R^2 \sim O(1/L) \rightarrow 0$.
2. **Multinomial contamination.** The OLS fit to $\log P(\mathbf{x})$ captures $\mathbb{E}[\log P(\mathbf{x}) \mid \mathbf{f}]$, which includes the effect of the multinomial coefficient $M(\mathbf{f}) = L! / \prod_j (L f_j)!$. By Stirling's approximation, $\log M(\mathbf{f}) \approx L H(\mathbf{f})$, where $H(\mathbf{f}) = -\sum_j f_j \log f_j$ is the entropy of $\mathbf{f}$. This contributes a quadratic term of the same $O(L)$ scaling as the Gaussian curvature, so the two effects never separate.

3 Results

All experiments use the Mess3 process with uniform stationary distribution $\mu = (1/3, 1/3, 1/3)$.

3.1 Sequence-level fit

The R^2 drops monotonically, confirming that the quadratic in $\delta \mathbf{f}$ captures a vanishing fraction of the variance in $\log P(\mathbf{x})$. The ratio ≈ -3 (close to $-d_{\text{vocab}}$) reflects the multinomial coefficient contributing a comparable quadratic curvature.

Table 1: Quadratic fit of $\log P(\mathbf{x})$ on $\delta \mathbf{f}$. The R^2 decreases with L as expected.

L	N	R^2	Ratio $-J_{\text{fit}}/\Sigma_{\text{emp}}^{-1}$
15	50,000	0.607	-3.45
20	10,000	0.482	-3.34
30	10,000	0.344	-3.26
50	10,000	0.209	-3.00
100	50,000	0.107	-2.91
200	100,000	0.050	-2.77

Table 2: Histogram-based type-level analysis. We compare two estimates of the L -independent matrix $\Sigma_{\text{true}}^{-1}$: from the quadratic fit to $\log P(\text{type})$ (in CLT-scaled coordinates $\mathbf{z} = \sqrt{L} \delta \mathbf{f}$), and from $L \cdot \Sigma_{\text{emp}}$. Reported: the $(0,0)$ entry of each matrix, their ratio, and the relative Frobenius error.

L	N	Types	Med. count	R^2	$\Sigma_{\text{truefit}}^{-1}[0,0]$	$\Sigma_{\text{trueemp}}^{-1}[0,0]$	Frob. err.
15	50,000	136	351	0.302	-0.29	1.18	1.24
30	10,000	495	22	0.325	0.26	0.94	0.72
50	10,000	1,138	8	0.600	0.47	0.88	0.49
100	50,000	3,390	9	0.840	0.61	0.82	0.25
200	100,000	7,740	7	0.868	0.65	0.79	0.19
300	100,000	11,008	5	0.828	0.62	0.78	0.21

3.2 Type-level histogram fit

The histogram R^2 improves from 0.30 to 0.87 and the two estimates of $\Sigma_{\text{true}}^{-1}$ approach each other (Frobenius error decreasing from 1.24 to 0.19). However, the convergence stalls around $L = 200$ –300 because the median count per type drops below ~ 5 , introducing systematic histogram bias: $\mathbb{E}[\log(\text{count})] < \log \mathbb{E}[\text{count}]$ (Jensen’s inequality).

The number of distinct types grows as $\binom{L+d-1}{d-1} \approx L^{d-1}/2$, so maintaining a fixed number of samples per type requires $N \propto L^{d-1}$. For Mess3 ($d = 3$), this means $N \propto L^2$.

3.3 Convergence of $\Sigma_{\text{true}}^{-1}$ estimates

The empirical estimate $\Sigma_{\text{trueemp}}^{-1} = (L \cdot \Sigma_{\text{emp}})^{-1}$ converges from above, while the histogram fit estimate $\Sigma_{\text{truefit}}^{-1}$ converges from below (due to Jensen bias). As of $L = 200$, the relative gap between them is approximately 20%.

Extrapolating, the true $\Sigma_{\text{true}}^{-1}$ for Mess3 (projected to the first two components) appears to be approximately

$$\Sigma_{\text{true}}^{-1} \approx \begin{pmatrix} 0.75 & 0.37 \\ 0.37 & 0.75 \end{pmatrix}, \quad (6)$$

which is consistent with both estimates at the largest L values.

4 Discussion

1. The CLT for HMM token frequencies is clearly operative: the empirical covariance Σ_{emp} scales as $1/L$ (i.e., $L \cdot \Sigma_{\text{emp}}$ stabilizes), and the type distribution is increasingly well-described by a

Gaussian.

2. The sequence-level log-likelihood $\log P(\mathbf{x})$ is *not* well-approximated by a quadratic in $\delta \mathbf{f}$ for large L , because ordering effects dominate. The useful Gaussian structure lives at the *type* level, not the sequence level.
3. The histogram-based type-level fit successfully recovers $\Sigma_{\text{true}}^{-1}$ but requires $N \propto L^2$ samples to avoid Jensen bias in the log-count estimator. This makes pushing to very large L expensive.
4. The forward algorithm is not needed for the type-level analysis: the histogram $P(\text{type } \mathbf{f}) = \text{count}_{\mathbf{f}}/N$ is a simpler and unbiased estimator. The forward algorithm remains useful for sequence-level diagnostics and for computing the observation entropy rate $h_{\text{obs}} = -\lim_{L \rightarrow \infty} \mathbb{E}[\log P(\mathbf{x})]/L$.